A Soft Procrustes Formula for Gaussian KL-Unbalanced Optimal Transport

Relaxed covariance factors, Riccati equations, and the Bures limit

Gabriel Peyré

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Abstract

Quadratic Wasserstein transport between Gaussian measures decomposes into a Euclidean cost between means and the Bures cost between covariances. The latter is exactly an orthogonal Procrustes problem between matrix square roots. This note studies the corresponding question for quadratic unbalanced optimal transport with Kullback–Leibler penalties on the two marginals and no entropy penalty on the coupling. After separating transported mass, the Gaussian problem is an infimal convolution of Wasserstein distance with two Gaussian KL divergences. We show that its covariance term has an exact *soft Procrustes* formulation: the hard Gram constraints of Bures are relaxed by LogDet penalties, or equivalently the orthogonal factors are replaced by general linear factors penalized for departing from the orthogonal group. A single orthogonal Procrustes alignment remains after fixing two positive stretches, but finite KL strength does not reduce to one hard orthogonal alignment. We connect this representation to the known Riccati closed form, treat unequal marginal penalties, recover ordinary Bures–Procrustes as the large-penalty limit, and clarify which convexity properties survive.

1 Question and answer

The balanced Gaussian formula suggests asking whether the KL-relaxed problem also comes from aligning covariance factors. The answer is positive, with one important qualification.

Executive answer. Gaussian KL-unbalanced quadratic transport admits an exact Procrustes-type formula, but it is a *soft* rather than a hard Procrustes problem. If $\Sigma_i^{1/2}$ are prescribed covariance roots, the factors $\Sigma_i^{1/2}X_i$ are free and the departure of X_i from orthogonality is penalized by

$$\Psi(X_i) = \|X_i\|_{\mathrm{F}}^2 - \log\det(X_iX_i^\top) - d.$$

This penalty is non-negative and vanishes exactly on the orthogonal group. For fixed positive stretches, the remaining rotation is the usual one-SVD Procrustes alignment. In general, however, the stretches must also be optimized, so there is no reduction to a single hard orthogonal alignment. The optimizer is nevertheless explicit through the same Riccati equation as the Gaussian KL-UOT closed form.

The strictly entropic problem, which adds $\epsilon\,\mathrm{KL}(\pi \mid \alpha\otimes\beta)$ to the coupling objective, is not the object considered here. Its Gaussian solution is also explicit (Janati et al., 2020), but the optimal joint Gaussian is non-degenerate. Here there is no coupling entropy: the optimizer is a scaled Wasserstein graph coupling between two adjusted Gaussian marginals.

The symmetric endpoint can be recovered from the Gaussian formulas of Janati et al. (2020); the two independent penalties are treated directly by Yang and Zhang (2026).

2 KL-unbalanced quadratic transport

For finite non-negative measures ρ and η, use the generalized relative entropy

$$\mathrm{KL}(\rho \mid \eta) = \begin{cases} \int (r \log r - r + 1)\, \mathrm{d}\eta, & \rho = r\eta, \\ +\infty, & \rho \not\ll \eta. \end{cases} \tag{1}$$

For a common marginal penalty $\tau > 0$, define

$$\mathsf{U}_\tau(\alpha_0, \alpha_1) = \inf_{\pi \in \mathcal{M}_+(\mathbb{R}^d \times \mathbb{R}^d)} \left\{ \int \|x - y\|^2 \, \mathrm{d}\pi(x, y) + \tau\, \mathrm{KL}(\pi_0 \mid \alpha_0) + \tau\, \mathrm{KL}(\pi_1 \mid \alpha_1) \right\}. \tag{2}$$

With the normalization in (2), $\mathsf{U}_\tau^{1/2}$ is the Gaussian Hellinger–Kantorovich endpoint metric on finite measures over Euclidean space. It is related to, but should not be confused with, the intrinsic Wasserstein–Fisher–Rao length metric; see Liero et al. (2018). This note only studies the static cost (2); see also Chizat et al. (2018) for the broader unbalanced transport framework.

2.1 Mass–shape separation

Let $\alpha_i = a_i \mu_i$, where $a_i > 0$ and μ_i are probability measures. Introduce the normalized shape envelope

$$\mathcal{E}_\tau(\mu_0, \mu_1) = \inf_{\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ W_2^2(\rho_0, \rho_1) + \tau\, \mathrm{KL}(\rho_0 \mid \mu_0) + \tau\, \mathrm{KL}(\rho_1 \mid \mu_1) \right\}. \tag{3}$$

Proposition 2.1 (Mass–shape decomposition)**.** *For probability measures μ_0, μ_1 for which $\mathcal{E}_\tau(\mu_0, \mu_1) < \infty$,*

$$\mathsf{U}_\tau(a_0\mu_0, a_1\mu_1) = \tau \left[a_0 + a_1 - 2\sqrt{a_0 a_1} \exp\left(-\frac{\mathcal{E}_\tau(\mu_0, \mu_1)}{2\tau} \right) \right]. \tag{4}$$

If (ρ_0^*, ρ_1^*) minimizes (3), the optimal transported mass is

$$M^* = \sqrt{a_0 a_1} \exp\left(-\frac{\mathcal{E}_\tau(\mu_0, \mu_1)}{2\tau} \right). \tag{5}$$

The optimal plan is M^* times a W_2-optimal probability coupling between ρ_0^* and ρ_1^*.

Proof. Write a nonzero plan as $\pi = M\bar{\pi}$, where M is its total mass and $\bar{\pi}$ is a probability coupling with marginals ρ_0, ρ_1. The homogeneity identity

$$\mathrm{KL}(M\rho_i \mid a_i\mu_i) = M\, \mathrm{KL}(\rho_i \mid \mu_i) + M \log(M/a_i) - M + a_i$$

reduces the objective, after minimizing $\bar{\pi}$ at fixed marginals, to

$$\begin{aligned} M\mathcal{A}(\rho_0, \rho_1) &+ \tau\{M \log(M/a_0) - M + a_0\} \\ &+ \tau\{M \log(M/a_1) - M + a_1\}, \end{aligned}$$

where $\mathcal{A}$ is the expression minimized in (3). Differentiation in M gives $M = \sqrt{a_0 a_1} e^{-\mathcal{A}/(2\tau)}$. Substitution gives $\tau(a_0 + a_1 - 2M)$, which is increasing in $\mathcal{A}$. Minimizing the remaining shape variables proves both claims. □

The nonlinear exponential in (4) is essential. The shape envelope $\mathcal{E}_\tau$ is not itself the squared unbalanced distance; it is the energy whose exponential controls how much mass is worth transporting.

3 Exact Gaussian reduction

Assume from now on that

$$\mu_i = \mathcal{N}(m_i, \Sigma_i), \qquad m_i \in \mathbb{R}^d, \quad \Sigma_i \in \mathbb{S}^d_{++}. \tag{6}$$

For $P, \Sigma \in \mathbb{S}^d_{++}$, define the LogDet divergence

$$D_{\mathrm{ld}}(P \mid \Sigma) = \operatorname{tr}(\Sigma^{-1}P) - \log\det(\Sigma^{-1}P) - d. \tag{7}$$

The Gaussian KL formula reads

$$\mathrm{KL}(\mathcal{N}(u,P) \mid \mathcal{N}(m,\Sigma)) = \frac{1}{2}\left(\|u-m\|^2_{\Sigma^{-1}} + D_{\mathrm{ld}}(P \mid \Sigma)\right). \tag{8}$$

Proposition 3.1 (Gaussian reduction)**.** *The infimum in (3) is attained by Gaussian measures. More precisely,*

$$\begin{aligned}\mathcal{E}_\tau(\mu_0,\mu_1) = \min_{\substack{u_0,u_1\in\mathbb{R}^d\\ P_0,P_1\in\mathbb{S}^d_{++}}} \Big\{ &\|u_0-u_1\|^2 + \mathsf{B}^2(P_0,P_1) \\ &+ \frac{\tau}{2}\left(\|u_0-m_0\|^2_{\Sigma_0^{-1}} + D_{\mathrm{ld}}(P_0 \mid \Sigma_0)\right) \\ &+ \frac{\tau}{2}\left(\|u_1-m_1\|^2_{\Sigma_1^{-1}} + D_{\mathrm{ld}}(P_1 \mid \Sigma_1)\right)\Big\},\end{aligned} \tag{9}$$

where

$$\mathsf{B}^2(P_0,P_1) = \operatorname{tr} P_0 + \operatorname{tr} P_1 - 2\operatorname{tr}\left(P_0^{1/2}P_1P_0^{1/2}\right)^{1/2}. \tag{10}$$

Proof. Let ρ_i have mean u_i and covariance P_i. Gelbrich's inequality gives

$$W_2^2(\rho_0,\rho_1) \geq W_2^2(\mathcal{N}(u_0,P_0), \mathcal{N}(u_1,P_1)).$$

Moreover, because the logarithm of a ratio of Gaussian densities is quadratic, moment matching gives the Pythagorean identity

$$\mathrm{KL}(\rho_i \mid \mu_i) = \mathrm{KL}(\rho_i \mid \mathcal{N}(u_i,P_i)) + \mathrm{KL}(\mathcal{N}(u_i,P_i) \mid \mu_i).$$

Replacing each ρ_i by its moment-matching Gaussian can therefore only decrease the objective. The Gaussian W_2 and KL formulas then give (9). This argument is the Gaussian reduction used in Yang and Zhang (2026); see also Gelbrich (1990). □

The mean and covariance variables in (9) separate. Writing $\Delta m = m_0 - m_1$, direct quadratic minimization gives

$$\mathcal{E}_{\tau,\mathrm{mean}} = \frac{\tau}{2}\Delta m^\top \left(\Sigma_0 + \Sigma_1 + \frac{\tau}{2}\mathrm{Id}\right)^{-1}\Delta m. \tag{11}$$

The nontrivial part is therefore the covariance envelope

$$\mathcal{C}_\tau(\Sigma_0,\Sigma_1) = \min_{P_0,P_1\in\mathbb{S}^d_{++}} \left\{\mathsf{B}^2(P_0,P_1) + \frac{\tau}{2}D_{\mathrm{ld}}(P_0 \mid \Sigma_0) + \frac{\tau}{2}D_{\mathrm{ld}}(P_1 \mid \Sigma_1)\right\}. \tag{12}$$

4 The soft Procrustes representation

Classical Bures geometry permits two equivalent factor formulations (Bhatia et al., 2019; Masarotto et al., 2019). For $d' \geq d$,

$$\mathsf{B}^2(P_0, P_1) = \min_{\substack{F_i \in \mathbb{R}^{d \times d'} \\ F_i F_i^\top = P_i}} \|F_0 - F_1\|_{\mathrm{F}}^2 . \tag{13}$$

If fixed factors $A_i A_i^\top = P_i$ are used instead, this is the orthogonal Procrustes formula

$$\mathsf{B}^2(P_0, P_1) = \min_{R \in \mathrm{O}(d')} \|A_0 - A_1 R\|_{\mathrm{F}}^2 . \tag{14}$$

The KL-relaxed problem preserves (13), but replaces the hard Gram constraints by penalties.

For a full-row-rank matrix $X \in \mathbb{R}^{d \times d'}$, define

$$\Psi(X) = \|X\|_{\mathrm{F}}^2 - \log\det(XX^\top) - d. \tag{15}$$

Theorem 4.1 (Soft Procrustes formula)**.** *Let $S_i = \Sigma_i^{1/2}$ and $d' \geq d$. Then*

$$\begin{aligned} \mathcal{C}_\tau(\Sigma_0, \Sigma_1) = \min_{\substack{X_i \in \mathbb{R}^{d \times d'} \\ \operatorname{rank} X_i = d}} \Big\{ & \|S_0 X_0 - S_1 X_1\|_{\mathrm{F}}^2 \\ & + \frac{\tau}{2}\Psi(X_0) + \frac{\tau}{2}\Psi(X_1) \Big\}. \end{aligned} \tag{16}$$

The penalty Ψ is non-negative and

$$\Psi(X) = 0 \quad \Longleftrightarrow \quad XX^\top = \mathrm{Id}. \tag{17}$$

Thus the hard row-Stiefel constraints of Bures–Procrustes are softened by LogDet penalties. For $d' = d$, their zero set is exactly $\mathrm{O}(d)$.

Proof. Insert the free-factor identity (13) into (12) and minimize jointly over F_0, F_1. Every full-row-rank factor can be written $F_i = S_i X_i$. Cyclicity of the trace and similarity invariance of the determinant give

$$\begin{aligned} D_{\mathrm{ld}}(F_i F_i^\top \mid \Sigma_i) &= \operatorname{tr}(X_i X_i^\top) - \log\det(X_i X_i^\top) - d \\ &= \Psi(X_i), \end{aligned}$$

which proves (16). If $s_1, \ldots, s_d$ are the singular values of X, then

$$\Psi(X) = \sum_{j=1}^{d} (s_j^2 - 2\log s_j - 1).$$

Each summand is non-negative and vanishes only at $s_j = 1$, proving (17). □

This identity can also be read directly in the original factor variables:

$$\mathcal{C}_\tau = \min_{F_0, F_1} \left\{ \|F_0 - F_1\|_{\mathrm{F}}^2 + \frac{\tau}{2} D_{\mathrm{ld}}(F_0 F_0^\top \mid \Sigma_0) + \frac{\tau}{2} D_{\mathrm{ld}}(F_1 F_1^\top \mid \Sigma_1) \right\}. \tag{18}$$

Ordinary Bures imposes $F_i F_i^\top = \Sigma_i$ exactly; KL-unbalanced Bures penalizes violation of these two Gram constraints.

4.1 One classical Procrustes step remains

The soft formula is not a single orthogonal alignment, but it contains one. Set $d' = d$ and use the left polar decompositions $X_i = H_i U_i$, with $H_i \succ 0$ and $U_i \in \mathrm{O}(d)$. The objective is invariant under simultaneous right multiplication of (X_0, X_1) by an orthogonal matrix. Gauging out U_0 gives the following equivalent form.

Corollary 4.2 (Stretched Procrustes formula)**.** *Define*

$$\psi(H) = \mathrm{tr}(H^2) - 2\log\det H - d, \qquad H \in \mathbb{S}^d_{++}. \tag{19}$$

Then

$$\begin{aligned}\mathcal{C}_\tau(\Sigma_0, \Sigma_1) = \min_{\substack{H_0, H_1 \in \mathbb{S}^d_{++} \\ R \in \mathrm{O}(d)}} \Big\{ &\|S_0 H_0 - S_1 H_1 R\|_{\mathrm{F}}^2 \\ &+ \frac{\tau}{2}\psi(H_0) + \frac{\tau}{2}\psi(H_1)\Big\}. \end{aligned} \tag{20}$$

For fixed H_0, H_1, the minimizer in R is the ordinary orthogonal Procrustes solution. Consequently,

$$\begin{aligned}\mathcal{C}_\tau = \min_{H_0, H_1 \in \mathbb{S}^d_{++}} \Big\{ &\mathrm{tr}(\Sigma_0 H_0^2) + \mathrm{tr}(\Sigma_1 H_1^2) - 2\left\|(S_1 H_1)^\top (S_0 H_0)\right\|_* \\ &+ \frac{\tau}{2}\psi(H_0) + \frac{\tau}{2}\psi(H_1)\Big\}, \end{aligned} \tag{21}$$

where $\|\cdot\|_*$ is the nuclear norm.

Proof. Right-multiply both polar factors by $U_0^\top$. This leaves all Frobenius norms and both penalties unchanged, while replacing (X_0, X_1) by $(H_0, H_1 R)$ with $R = U_1 U_0^\top$. The first identity follows. For fixed stretches, expanding the squared norm reduces the problem to maximizing a trace over $R \in \mathrm{O}(d)$. The von Neumann trace inequality gives the nuclear norm in (21); the maximizing rotation is obtained from one singular-value decomposition (Schönemann, 1966). □

The two positive stretches are the essential new degrees of freedom. Since ψ is not quadratic, the orbit argument behind (14) does not eliminate them. Thus (20) is an exact generalized Procrustes formula, but not a one-SVD endpoint formula.

4.2 Means, covariances, and masses in one formula

The entire Gaussian problem admits the same interpretation. For $\xi_i \in \mathbb{R}^d$ and $X_i \in \mathrm{GL}(d)$, define

$$\begin{aligned}\mathcal{J}_\tau(\xi_0, \xi_1, X_0, X_1) = {} &\|\Delta m + S_0\xi_0 - S_1\xi_1\|^2 + \|S_0 X_0 - S_1 X_1\|_{\mathrm{F}}^2 \\ &+ \frac{\tau}{2}\left(\|\xi_0\|^2 + \|\xi_1\|^2 + \Psi(X_0) + \Psi(X_1)\right). \end{aligned} \tag{22}$$

Corollary 4.3 (Gaussian KL-UOT as exponential soft Procrustes)**.** *For $\alpha_i = a_i\mathcal{N}(m_i, \Sigma_i)$,*

$$\begin{aligned}\mathsf{U}_\tau(\alpha_0, \alpha_1) = \min_{\xi_0, \xi_1, X_0, X_1} \tau\Big[&a_0 + a_1 - 2\sqrt{a_0 a_1} \\ &\times \exp\Big(-\frac{\mathcal{J}_\tau(\xi_0, \xi_1, X_0, X_1)}{2\tau}\Big)\Big]. \end{aligned} \tag{23}$$

The minimization is over $\xi_i \in \mathbb{R}^d$ and $X_i \in \mathrm{GL}(d)$.

Proof. Write $u_i = m_i + S_i\xi_i$ in (9), use theorem 4.1, and then apply proposition 2.1. The scalar function $s \mapsto \tau(a_0 + a_1 - 2\sqrt{a_0a_1}e^{-s/(2\tau)})$ is increasing, so the minimum can be placed outside as in (23). $\square$

For centered unit-mass inputs, the restriction

$$\mathsf{UB}_\tau^2(\Sigma_0, \Sigma_1) = \mathsf{U}_\tau(\mathcal{N}(0, \Sigma_0), \mathcal{N}(0, \Sigma_1)) = 2\tau\left(1 - e^{-\mathcal{C}_\tau/(2\tau)}\right) \tag{24}$$

is the natural KL-unbalanced or Gaussian-Hellinger generalization of the Bures metric. Formula (16) is the requested Procrustes-type representation, with the nonlinear conversion (24) from normalized shape energy to the actual squared metric.

5 Closed-form optimizer and the hidden Bures step

The soft Procrustes problem is not merely a conceptual reformulation. Its optimizer is the adjusted covariance pair appearing in the Gaussian KL-UOT closed form. This section records the equal-penalty formula; the asymmetric extension is given in section 7.

Set

$$\eta = \frac{2}{\tau}, \qquad A_i = \Sigma_i^{-1}, \qquad C_i = A_i + \eta\mathrm{Id}. \tag{25}$$

Define

$$L_* = C_1^{-1/2}\left(C_1^{1/2}C_0C_1^{1/2}\right)^{1/2}C_1^{-1/2}, \tag{26}$$

and

$$P_* = [A_0 + \eta(\mathrm{Id} - L_*)]^{-1}, \qquad Q_* = L_*P_*L_*. \tag{27}$$

For the means, let

$$h_* = [\mathrm{Id} + \eta(\Sigma_0 + \Sigma_1)]^{-1}\Delta m, \quad u_0^* = m_0 - \eta\Sigma_0 h_*, \quad u_1^* = m_1 + \eta\Sigma_1 h_*. \tag{28}$$

Proposition 5.1 (Riccati solution of soft Procrustes)**.** *The matrices P_*, Q_* in (27) are positive definite and uniquely minimize (12). The normalized shape value is*

$$\begin{aligned}\mathcal{E}_\tau(\mu_0, \mu_1) = {}& \Delta m^\top[\mathrm{Id} + \eta(\Sigma_0 + \Sigma_1)]^{-1}\Delta m \\ & + \frac{\tau}{2}\log\frac{\det\Sigma_0\det\Sigma_1}{\det P_*\det Q_*}.\end{aligned} \tag{29}$$

One minimizing factor pair in (16) is

$$X_0^* = S_0^{-1}P_*^{1/2}, \qquad X_1^* = S_1^{-1}L_*P_*^{1/2}. \tag{30}$$

Proof. Let $L \succ 0$ be the Gaussian optimal map from P to Q, so $Q = LPL$. The differential of $\mathsf{B}^2(P, Q)$ with respect to P is $\mathrm{Id} - L$, and its differential with respect to Q is $\mathrm{Id} - L^{-1}$. The first-order equations for (12) are therefore

$$P^{-1} = A_0 + \eta(\mathrm{Id} - L), \qquad Q^{-1} = A_1 + \eta(\mathrm{Id} - L^{-1}). \tag{31}$$

Using $Q^{-1} = L^{-1}P^{-1}L^{-1}$ gives the Riccati equation

$$LC_1L = C_0. \tag{32}$$

Its unique positive-definite solution is (26), and back substitution gives (27). Global optimality and uniqueness follow either from the KL-UOT dual certificate in Yang and

Zhang (2026), or from joint convexity of the Bures term and strict convexity of the two LogDet terms; see proposition 6.2 below.

At stationarity, taking traces in (31) cancels the trace part of both Gaussian KL divergences against the Bures term. Only the displayed log-determinants remain. Combining them with the quadratic mean minimum gives (29). Finally, setting $F_0 = P_*^{1/2}$ and $F_1 = L_* P_*^{1/2}$ gives $F_i F_i^\top = P_i^*$ and

$$\|F_0 - F_1\|_{\mathrm{F}}^2 = \mathsf{B}^2(P_*, Q_*).$$

Writing $F_i = S_i X_i^*$ yields (30). □

Equation (32) reveals a second, internal Procrustes structure: L_* is exactly the Gaussian optimal map from the shifted precision matrix C_1 to C_0. Hence the finite-penalty solution does use an ordinary Bures square root or Procrustes alignment, but it applies to shifted precisions and must be followed by the back-substitution (27).

6 What survives from ordinary Bures geometry?

6.1 Large marginal penalties

The soft formulation makes the balanced limit transparent. Since $\Psi(X) \geq 0$ with equality exactly on the row-Stiefel manifold, increasing τ enforces $X_i X_i^\top \to \mathrm{Id}$ and $\xi_i \to 0$. In the square case this recovers the hard orthogonal factors of Bures–Procrustes.

Proposition 6.1 (Bures–Procrustes limit)**.** *For Gaussian probability measures $\mu_i = \mathcal{N}(m_i, \Sigma_i)$,*

$$\mathcal{E}_\tau(\mu_0, \mu_1) \longrightarrow W_2^2(\mu_0, \mu_1) \qquad (\tau \to \infty). \tag{33}$$

Consequently, if $a_0 = a_1 = a$,

$$\mathsf{U}_\tau(a\mu_0, a\mu_1) \longrightarrow a W_2^2(\mu_0, \mu_1). \tag{34}$$

If the masses differ, the leading term is $\tau(\sqrt{a_0} - \sqrt{a_1})^2$.

Proof. The upper bound in (33) follows by choosing $\rho_i = \mu_i$ in (3). Bounded shape energy forces the two KL terms to vanish as $\tau \to \infty$, hence any limit of minimizers has marginals μ_0, μ_1; lower semicontinuity gives the converse bound. The same conclusion follows directly from (22): bounded energy forces $\xi_i \to 0$ and $X_i X_i^\top \to \mathrm{Id}$, after which the remaining minimization is ordinary Gaussian Bures–Procrustes. Expanding the exponential in (4) proves the last two claims. □

6.2 Convexity: inner problem versus endpoint parameters

The word "Procrustes" often suggests a nonconvex matrix factorization. The natural covariance variables show that the adjusted-covariance problem is in fact convex.

Proposition 6.2 (Convexity of the adjusted-covariance problem)**.** *For fixed $\Sigma_0, \Sigma_1 \succ 0$, the objective in (12) is strictly jointly convex in (P_0, P_1). It therefore has a unique minimizer. The factor objective (16) is a redundant, nonconvex parameterization of this convex problem and is invariant under simultaneous right orthogonal actions.*

Proof. The squared Bures function is jointly convex on pairs of positive semidefinite matrices (Bhatia et al., 2019). For fixed Σ, the map $P \mapsto D_{\mathrm{ld}}(P \mid \Sigma)$ is strictly convex because its only nonlinear term is $-\log\det P$, whose Hessian is positive definite on symmetric directions. The sum is therefore strictly jointly convex. The factor invariance follows from $(F_0, F_1) \mapsto (F_0 R, F_1 R)$ for $R \in \mathrm{O}(d')$. □

This statement should not be confused with joint convexity in the prescribed covariances (Σ_0, Σ_1). Unlike squared Bures distance, the Gaussian restriction of KL-UOT is not jointly convex in these parameters, even in one dimension.

Example 6.3 (Failure of covariance joint convexity)**.** Take $d = 1$, centered unit-mass Gaussians and $\tau = 1$. Write $s_i > 0$ for the variance. Set

$$L = \sqrt{\frac{s_0^{-1} + 2}{s_1^{-1} + 2}}, \qquad P = [s_0^{-1} + 2(1 - L)]^{-1}, \qquad Q = L^2 P.$$

Then (24) becomes

$$\mathsf{UB}_1^2(s_0, s_1) = 2\left[1 - \left(\frac{PQ}{s_0 s_1}\right)^{1/4}\right]. \tag{35}$$

Direct substitution gives

$$\begin{aligned}
\mathsf{UB}_1^2(0.1, 0.02) &= 0.027134127639,\\
\mathsf{UB}_1^2(0.3, 0.04) &= 0.089114415591,\\
\mathsf{UB}_1^2(0.2, 0.03) &= 0.060423206395.
\end{aligned}$$

The last pair is the midpoint of the first two, but

$$0.060423206395 > \tfrac{1}{2}(0.027134127639 + 0.089114415591) = 0.058124271615.$$

Thus covariance joint convexity fails. There is no contradiction with joint convexity of KL-UOT in the input measures: a mixture of two Gaussian measures is generally not the Gaussian whose covariance is the arithmetic mean.

6.3 Why this is not one hard Procrustes problem

The balanced Bures formula optimizes over a compact gauge orbit while keeping the endpoint Gram matrices fixed. At finite τ, the optimal factors in (16) generally satisfy $X_i X_i^\top \neq \mathrm{Id}$: their singular values encode the KL-optimal covariance relaxation. The nonquadratic LogDet penalty prevents the same orbit argument from removing these stretches. One may compute the best rotation by one SVD for each pair (H_0, H_1), but the stretches remain coupled through (21).

This does not constitute a theorem excluding every conceivable finite-dimensional lift. It establishes the sharper statement relevant to the standard Bures derivation: the natural square-root lift is exact, but it is a penalized $\mathrm{GL}(d)$ alignment rather than a quotient by a single orthogonal orbit. We did not find this soft-Procrustes identity stated in the Gaussian KL-UOT references; it follows directly from their Gaussian reduction combined with the classical free-factor Bures formula.

7 Unequal marginal penalties

The same representation applies to the asymmetric problem

$$\mathsf{U}_{\tau_0,\tau_1}(\alpha_0,\alpha_1) = \inf_{\pi} \left\{ \int \|x-y\|^2 \, \mathrm{d}\pi + \tau_0 \, \mathrm{KL}(\pi_0 \mid \alpha_0) + \tau_1 \, \mathrm{KL}(\pi_1 \mid \alpha_1) \right\}. \tag{36}$$

This is no longer symmetric when $\tau_0 \neq \tau_1$, so it should be called a cost rather than a distance. Put $T = \tau_0 + \tau_1$ and $\theta_i = \tau_i / T$. Define

$$\begin{aligned} \mathcal{J}_{\tau_0,\tau_1} = \; & \|\Delta m + S_0\xi_0 - S_1\xi_1\|^2 + \|S_0X_0 - S_1X_1\|_{\mathrm{F}}^2 \\ & + \frac{\tau_0}{2}\{\|\xi_0\|^2 + \Psi(X_0)\} + \frac{\tau_1}{2}\{\|\xi_1\|^2 + \Psi(X_1)\}. \end{aligned} \tag{37}$$

The mass calculation in proposition 2.1 gives the exact asymmetric soft-Procrustes formula

$$\begin{aligned} \mathsf{U}_{\tau_0,\tau_1}(a_0\mu_0, a_1\mu_1) = \min_{\xi_i, X_i} \Big\{ & \tau_0 a_0 + \tau_1 a_1 \\ & - T a_0^{\theta_0} a_1^{\theta_1} \exp\Big(-\frac{\mathcal{J}_{\tau_0,\tau_1}}{T} \Big) \Big\}. \end{aligned} \tag{38}$$

For completeness, the closed form of Yang and Zhang (2026) can be recovered from this representation. Set

$$\eta_i = \frac{2}{\tau_i}, \qquad A_i = \Sigma_i^{-1}, \qquad C_i = A_i + \eta_i \mathrm{Id}, \qquad \kappa = \eta_1 - \eta_0. \tag{39}$$

Define

$$\begin{aligned} S_* &= \frac{1}{2}\left[\kappa \mathrm{Id} + \left(\kappa^2 \mathrm{Id} + 4 C_1^{1/2} C_0 C_1^{1/2} \right)^{1/2} \right], \\ L_* &= C_1^{-1/2} S_* C_1^{-1/2}, \end{aligned} \tag{40}$$

and

$$P_* = [A_0 + \eta_0(\mathrm{Id} - L_*)]^{-1}, \qquad Q_* = L_* P_* L_*. \tag{41}$$

Then L_* is the unique positive solution of

$$L_* C_1 L_* - \kappa L_* = C_0. \tag{42}$$

For the means,

$$h_* = [\mathrm{Id} + \eta_0\Sigma_0 + \eta_1\Sigma_1]^{-1}\Delta m, \quad u_* = m_0 - \eta_0\Sigma_0 h_*, \quad v_* = m_1 + \eta_1 \Sigma_1 h_*. \tag{43}$$

The minimum normalized energy is

$$\begin{aligned} \mathcal{E}_* = \; & \Delta m^\top [\mathrm{Id} + \eta_0\Sigma_0 + \eta_1\Sigma_1]^{-1}\Delta m \\ & + \frac{\tau_0}{2}\log\frac{\det \Sigma_0}{\det P_*} + \frac{\tau_1}{2}\log\frac{\det\Sigma_1}{\det Q_*}. \end{aligned} \tag{44}$$

The transported mass and final value are

$$M_* = a_0^{\theta_0} a_1^{\theta_1} e^{-\mathcal{E}_*/T}, \qquad \mathsf{U}_{\tau_0,\tau_1} = \tau_0 a_0 + \tau_1 a_1 - T M_*. \tag{45}$$

When $\tau_0 = \tau_1$, $\kappa = 0$ and (42) reduces to the internal Bures equation (32). Unequal penalties deform this equation by a linear Riccati term.

8 Conclusion

The KL-unbalanced Gaussian endpoint cost does have a precise analogue of the Bures–Procrustes formula. It is best understood through three nested layers:

1. total transported mass is eliminated analytically and produces the exponential conversion (4);
2. the normalized Gaussian problem chooses two adjusted means and covariances by KL penalization; and
3. the covariance choice is exactly a soft Procrustes alignment of free square-root factors, with LogDet penalties replacing hard Gram constraints.

At finite penalty, the correct symmetry is therefore not a larger hard orthogonal group. It is a simultaneous orthogonal gauge acting on two general linear factors whose singular values are optimized. One ordinary Procrustes step survives for fixed stretches, and the complete optimizer is available through a positive-definite Riccati equation. In the large-penalty limit, the stretches converge to orthogonal factors and the whole construction collapses to the classical Bures formula.

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